

DALLE-3

DALL·E로 이미지 생성하기

DALL·E 소개

- 프롬프트 기반의 이미지 생성 기본 예제
- 다른 크기의 이미지 생성하기
- 여러 이미지 생성하기
- 이미지 프롬프트 개선 방법
- 프롬프트 조합으로 다양한 랜덤 이미지 만들기

아래 이미지는 임시적으로 URL이 생성되고, 일정 시간이 지나면 사라지면 이미지 링크가 없어집니다. 샘플 코드에는 그래서 예전 실행 결과의 이미지가 보이지 않습니다.

- openai 설치
 - pip install openai

```
In [1]: # !pip install openai
        # !pip install httpx
```

```
In [2]: import os
        from openai import OpenAI
        from dotenv import load_dotenv
        load_dotenv(override=True)

        OPENAI_API_KEY = os.getenv("OPENAI_API_KEY")

        # OpenAI 클라이언트 초기화
        client = OpenAI()
```

```
kwargs = {  
    "prompt": "A beautiful landscape."  
}
```

이미지 생성

```
im = client.images.generate(**kwargs)  
print(im)
```

ImagesResponse(created=1768457844, background=None, data=[Image(b64_json=None, revised_prompt=None, url='https://oaidalleapiprodscus.blob.core.windows.net/private/org-oBzzAxWzofVRTZNu0vyqtv5g/user-3YLGCPD3BaT0GloeVuBPE3Y9/img-3WM9okqagSVMkmm6Za3wjeCb.png?st=2026-01-15T05%3A17%3A24Z&se=2026-01-15T07%3A17%3A24Z&sp=r&sv=2024-08-04&sr=b&rscd=inline&rsct=image/png&skoid=92b997d6-f0dd-439d-8087-b26c74d72365&sktid=a48cca56-e6da-484e-a814-9c849652bcb3&skt=2026-01-15T04%3A43%3A15Z&ske=2026-01-16T04%3A43%3A15Z&sk=b&skv=2024-08-04&sig=1i/8tdk2h8071Y1/8r1B73zp0bBAFA%2Brk9i8x5H4Mqc%3D')], output_format=None, quality=None, size=None, usage=None)

- 아래 코드는 만들어준 이미지를 다운 받아서 출력시키는 것으로 생성한 이미지를 예시로 보여드리기 위함입니다.

```
In [3]: from IPython.display import Image, display  
display(Image(filename="a_beautiful_landscape.png", width=512, height=512))
```



12.3 다른 크기의 이미지 생성하기

In [4]: `from IPython.display import Image, display`

```
prompt = "A beautiful landscape."
```

```
n = 1
```

```
size = "256x256"
```

```

kwargs = {
    "prompt" : prompt,
    "n" : n,
    "size" : size,
}

im = client.images.generate(**kwargs)

for i in range(n):
    print(im.data[i].url)
    display(Image(url=im.data[i].url))

```

<https://oaidalleapiprodscus.blob.core.windows.net/private/org-oBzzAxWzofVRTZNu0vyqtv5g/user-3YLGCPd3BaT0GloeVuBPE3Y9/img-rdqGeHrGDuHjQmXWQy3QuGgs.png?st=2026-01-15T05%3A17%3A35Z&se=2026-01-15T07%3A17%3A35Z&sp=r&sv=2024-08-04&sr=b&rscd=inline&rsct=image/png&skoid=1726b4ce-fee1-450b-8b92-1731ad8745f6&sktid=a48cca56-e6da-484e-a814-9c849652bcb3&skt=2026-01-15T06%3A17%3A35Z&ske=2026-01-16T06%3A17%3A35Z&sks=b&skv=2024-08-04&sig=krrn3Uhd58wi6hm%2Bet0bZBkbr5dl/CUzY5pYeEwMg2FM%3D>



12.4 여러 이미지 생성하기

```

In [5]: from IPython.display import Image, display

prompt = "A beautiful landscape."
n = 3
size = "512x512"
kwargs = {

```



```

    "prompt": prompt,
    "n": n,
    "size" : size,
}

im = client.images.generate(**kwargs)
print(im)

for i in range(n):
    print(im.data[i].url)
    display(Image(url=im.data[i].url))

```

ImagesResponse(created=1768457865, background=None, data=[Image(b64_json=None, revised_prompt=None, url='https://oaidalleapiprodscus.blob.core.windows.net/private/org-oBzzAxWzofVRTZNu0vyqtv5g/user-3YLGCPD3BaT0GloeVuBPE3Y9/img-vwvblCSJjgciIKPg8lKSLt3S.png?st=2026-01-15T05%3A17%3A45Z&se=2026-01-15T07%3A17%3A45Z&sp=r&sv=2024-08-04&sr=b&rscd=inline&rsct=image/png&skoid=75dd98d5-7f83-456f-ae8d-a58a7fba1b61&sktid=a48cca56-e6da-484e-a814-9c849652bcb3&skt=2026-01-15T06%3A17%3A45Z&ske=2026-01-16T06%3A17%3A45Z&sks=b&skv=2024-08-04&sig=N7tBjIo07rdM6ItHqNVhyhhiewkM2kLgKEHSX6NLD2Q%3D'), Image(b64_json=None, revised_prompt=None, url='https://oaidalleapiprodscus.blob.core.windows.net/private/org-oBzzAxWzofVRTZNu0vyqtv5g/user-3YLGCPD3BaT0GloeVuBPE3Y9/img-6jcvHM4cVfMZLSvaXF2458Hw.png?st=2026-01-15T05%3A17%3A45Z&se=2026-01-15T07%3A17%3A45Z&sp=r&sv=2024-08-04&sr=b&rscd=inline&rsct=image/png&skoid=75dd98d5-7f83-456f-ae8d-a58a7fba1b61&sktid=a48cca56-e6da-484e-a814-9c849652bcb3&skt=2026-01-15T06%3A17%3A45Z&ske=2026-01-16T06%3A17%3A45Z&sks=b&skv=2024-08-04&sig=aqEzaBQlFQ/HvsNW38Skyul/26tntXzeTGHCy10kYCC%3D'), Image(b64_json=None, revised_prompt=None, url='https://oaidalleapiprodscus.blob.core.windows.net/private/org-oBzzAxWzofVRTZNu0vyqtv5g/user-3YLGCPD3BaT0GloeVuBPE3Y9/img-jWUOWToMrN9L1NVK4Av06SFt.png?st=2026-01-15T05%3A17%3A45Z&se=2026-01-15T07%3A17%3A45Z&sp=r&sv=2024-08-04&sr=b&rscd=inline&rsct=image/png&skoid=75dd98d5-7f83-456f-ae8d-a58a7fba1b61&sktid=a48cca56-e6da-484e-a814-9c849652bcb3&skt=2026-01-15T06%3A17%3A45Z&ske=2026-01-16T06%3A17%3A45Z&sks=b&skv=2024-08-04&sig=CXxcRk4V1m4smB5j7E2uqJaKZHyPsDvMmNy3VUiNeRs%3D')], output_format=None, quality=None, size=None, usage=None)

https://oaidalleapiprodscus.blob.core.windows.net/private/org-oBzzAxWzofVRTZNu0vyqtv5g/user-3YLGCPD3BaT0GloeVuBPE3Y9/img-vwvblCSJjgciIKPg8lKSLt3S.png?st=2026-01-15T05%3A17%3A45Z&se=2026-01-15T07%3A17%3A45Z&sp=r&sv=2024-08-04&sr=b&rscd=inline&rsct=image/png&skoid=75dd98d5-7f83-456f-ae8d-a58a7fba1b61&sktid=a48cca56-e6da-484e-a814-9c849652bcb3&skt=2026-01-15T06%3A17%3A45Z&ske=2026-01-16T06%3A17%3A45Z&sks=b&skv=2024-08-04&sig=N7tBjIo07rdM6ItHqNVhyhhiewkM2kLgKEHSX6NLD2Q%3D



<https://oaidalleapiprodscus.blob.core.windows.net/private/org-oBzzAxWzofVRTZNu0vyqtv5g/user-3YLGCPD3BaT0GloeVuBPE3Y9/img-6jcvHM4cVfMZLSvaXF2458Hw.png?st=2026-01-15T05%3A17%3A45Z&se=2026-01-15T07%3A17%3A45Z&sp=r&sv=2024-08-04&sr=b&rscd=inline&rsct=image/png&skoid=75dd98d5-7f83-456f-ae8d-a58a7fba1b61&sktid=a48cca56-e6da-484e-a814-9c849652bcb3&skt=2026-01-15T06%3A17%3A45Z&ske=2026-01-16T06%3A17%3A45Z&sks=b&skv=2024-08-04&sig=aqEzaBQlFQ/HvsNW38Skyul/26tntXzeTGHCy10kYCc%3D>



<https://oaidalleapiprodscus.blob.core.windows.net/private/org-oBzzAxWzofVRTZNu0vyqtv5g/user-3YLGCPD3BaT0GloeVuBPE3Y9/img-jwUOWToMrN9L1NVK4Av06SFt.png?st=2026-01-15T05%3A17%3A45Z&se=2026-01-15T07%3A17%3A45Z&sp=r&sv=2024-08-04&sr=b&rscd=inline&rsct=image/png&skoid=75dd98d5-7f83-456f-ae8d-a58a7fba1b61&sktid=a48cca56-e6da-484e-a814-9c849652bcb3&skt=2026-01-15T06%3A17%3A45Z&ske=2026-01-16T06%3A17%3A45Z&sks=b&skv=2024-08-04&sig=CXxcRk4V1m4smB5j7E2uqJaKZHyPsDvMmNy3VUiNeRs%3D>



12.5 이미지 프롬프트 개선 방법

작가 모방하기

```
In [6]: from IPython.display import Image, display
```

```

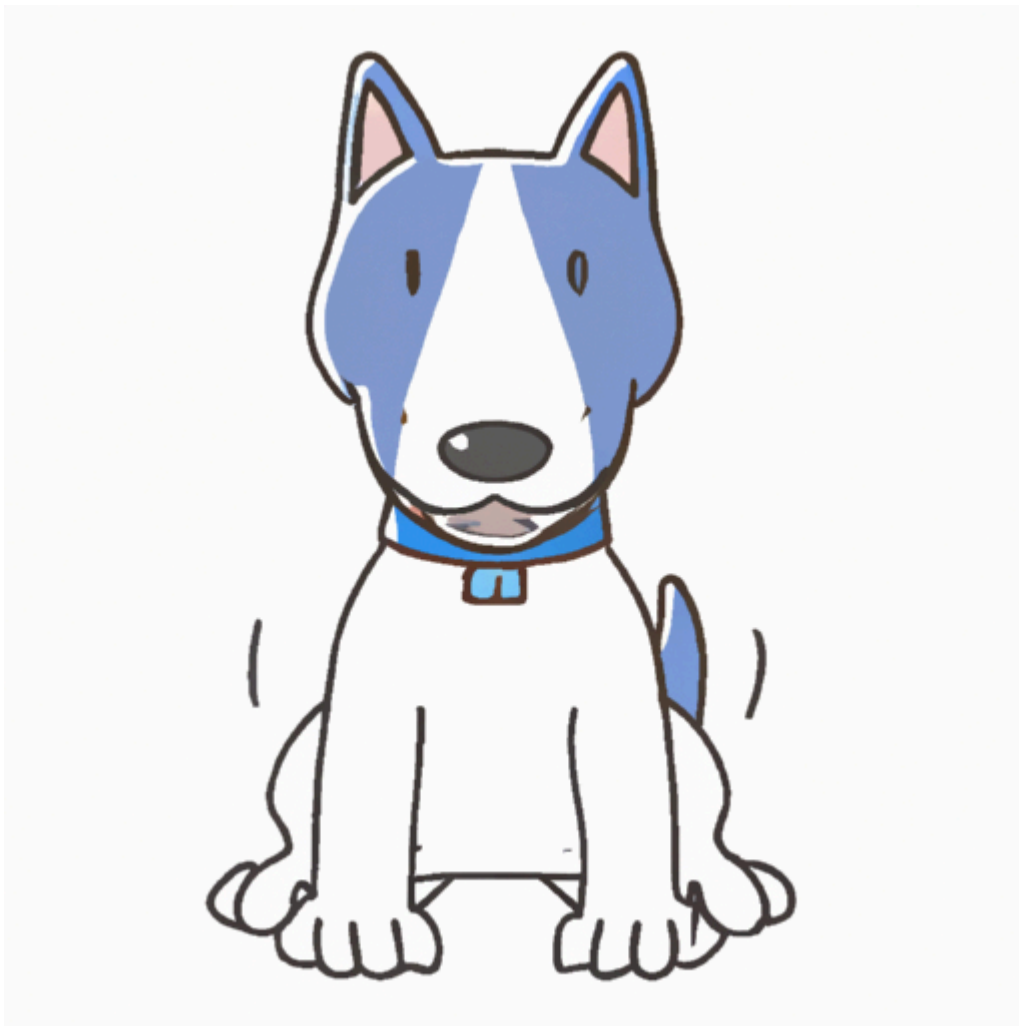
prompt = "cute smiling dog by Akira Toriyama"
n = 1
size = "512x512"
kwargs = {
    "prompt": prompt,
    "n": n,
    "size": size
}

im = client.images.generate(**kwargs)
print(im)

for i in range(n):
    print(im.data[i].url)
    display(Image(url=im.data[i].url))

```

ImagesResponse(created=1768457876, background=None, data=[Image(b64_json=None, revised_prompt=None, url='https://oaidalleapiprodscus.blob.core.windows.net/private/org-oBzzAxWzofVRTZNu0vyqtv5g/user-3YLGCPD3BaT0GloeVuBPE3Y9/img-dXDPDsGjppFpX8sQsGJAjBWR.png?st=2026-01-15T05%3A17%3A55Z&se=2026-01-15T07%3A17%3A55Z&sp=r&sv=2024-08-04&sr=b&rscd=inline&rsct=image/png&skoid=c4071b20-fbe5-4cbe-8e48-4476b70c6a90&sktid=a48cca56-e6da-484e-a814-9c849652bcb3&skt=2026-01-15T06%3A17%3A55Z&ske=2026-01-16T06%3A17%3A55Z&sks=b&skv=2024-08-04&sig=Afdx24bd4PB7IiYK5D4IM1TLsx7ALRU448v5f2I1Qs4%3D')], output_format=None, quality=None, size=None, usage=None)
 https://oaidalleapiprodscus.blob.core.windows.net/private/org-oBzzAxWzofVRTZNu0vyqtv5g/user-3YLGCPD3BaT0GloeVuBPE3Y9/img-dXDPDsGjppFpX8sQsGJAjBWR.png?st=2026-01-15T05%3A17%3A55Z&se=2026-01-15T07%3A17%3A55Z&sp=r&sv=2024-08-04&sr=b&rscd=inline&rsct=image/png&skoid=c4071b20-fbe5-4cbe-8e48-4476b70c6a90&sktid=a48cca56-e6da-484e-a814-9c849652bcb3&skt=2026-01-15T06%3A17%3A55Z&ske=2026-01-16T06%3A17%3A55Z&sks=b&skv=2024-08-04&sig=Afdx24bd4PB7IiYK5D4IM1TLsx7ALRU448v5f2I1Qs4%3D



```
In [7]: from IPython.display import Image, display
```

```
prompt = "평화로운 분위기의 아름다운 풍경"  
n = 1  
size = "512x512"  
kwargs = {  
    "prompt": prompt,  
    "n": n,
```



```

    "size": size
}

im = client.images.generate(**kwargs)
print(im)

for i in range(n):
    print(im.data[i].url)
    display(Image(url=im.data[i].url))

```

ImagesResponse(created=1768457889, background=None, data=[Image(b64_json=None, revised_prompt=None, url='https://oaidalleapiprodscus.blob.core.windows.net/private/org-oBzzAxWzofVRTZNu0vyqtv5g/user-3YLGCPD3BaT0GloeVuBPE3Y9/img-kx1L3qlwEvTNRjLNrigrbC3vH.png?st=2026-01-15T05%3A18%3A08Z&se=2026-01-15T07%3A18%3A08Z&sp=r&sv=2024-08-04&sr=b&rscd=inline&rsct=image/png&skoid=52f8f7b3-ca8d-4b21-9807-8b9df114d84c&sktid=a48cca56-e6da-484e-a814-9c849652bcb3&skt=2026-01-15T06%3A18%3A08Z&ske=2026-01-16T06%3A18%3A08Z&sks=b&skv=2024-08-04&sig=hosmJvibzE6TnMFJptTD4b0TSZLiMWsm/fD8gar5itk%3D')], output_format=None, quality=None, size=None, usage=None)

<https://oaidalleapiprodscus.blob.core.windows.net/private/org-oBzzAxWzofVRTZNu0vyqtv5g/user-3YLGCPD3BaT0GloeVuBPE3Y9/img-kx1L3qlwEvTNRjLNrigrbC3vH.png?st=2026-01-15T05%3A18%3A08Z&se=2026-01-15T07%3A18%3A08Z&sp=r&sv=2024-08-04&sr=b&rscd=inline&rsct=image/png&skoid=52f8f7b3-ca8d-4b21-9807-8b9df114d84c&sktid=a48cca56-e6da-484e-a814-9c849652bcb3&skt=2026-01-15T06%3A18%3A08Z&ske=2026-01-16T06%3A18%3A08Z&sks=b&skv=2024-08-04&sig=hosmJvibzE6TnMFJptTD4b0TSZLiMWsm/fD8gar5itk%3D>



```
In [8]: import os
        from openai import OpenAI
        from IPython.display import Image, display

        prompt = "beautiful landscape with a red sky"
        n = 1
        size = "512x512"
        kwargs = {
```

```

    "prompt": prompt,
    "n": n,
    "size": size
}

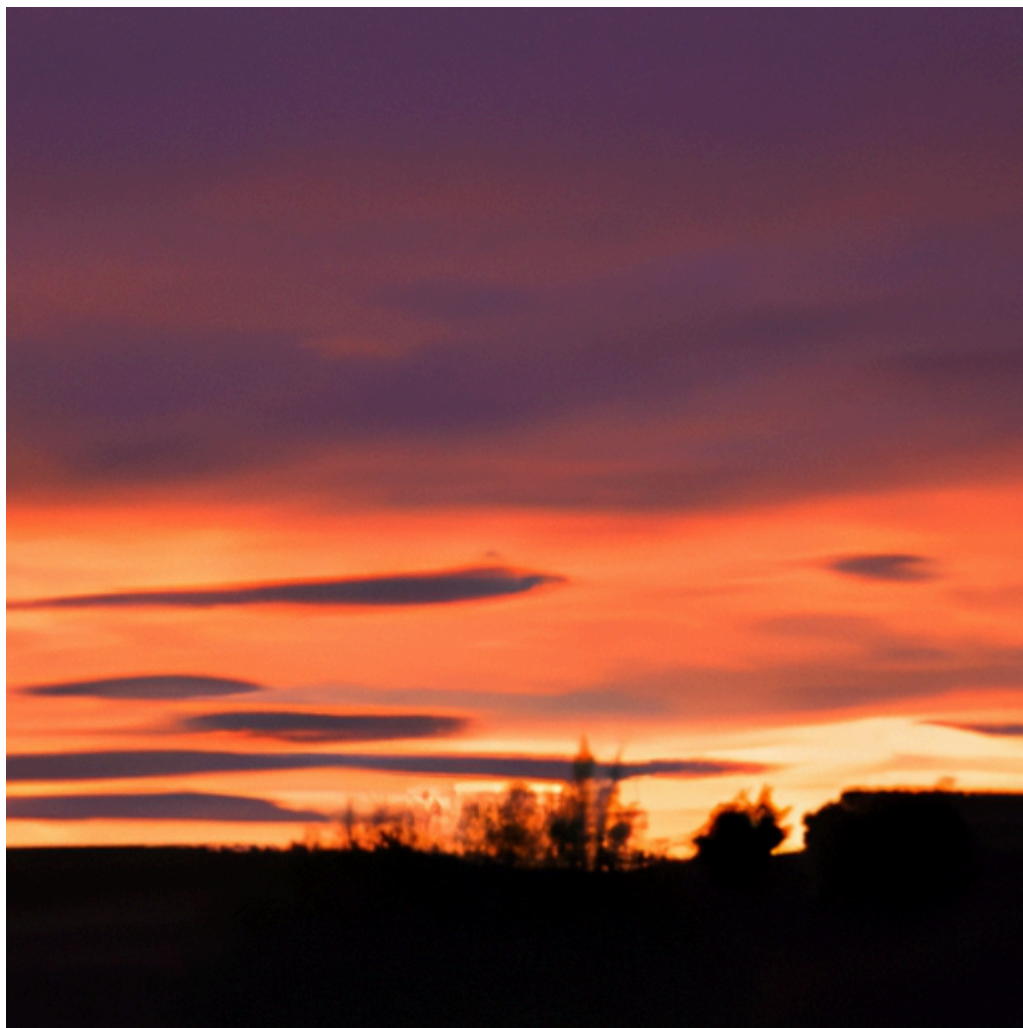
im = client.images.generate(**kwargs)
print(im)

for i in range(n):
    print(im.data[i].url)
    display(Image(url=im.data[i].url))

```

ImagesResponse(created=1768457899, background=None, data=[Image(b64_json=None, revised_prompt=None, url='https://oaidalleapiprodscus.blob.core.windows.net/private/org-oBzzAxWzofVRTZNu0vyqtv5g/user-3YLGCPD3BaT0GloeVuBPE3Y9/img-JQ9wMHFwAJG2bIn3ImD0orv0.png?st=2026-01-15T05%3A18%3A19Z&se=2026-01-15T07%3A18%3A19Z&sp=r&sv=2024-08-04&sr=b&rscd=inline&rsct=image/png&skoid=0ec745ad-3d9b-4d43-86db-87a2c9b420f1&sktid=a48cca56-e6da-484e-a814-9c849652bcb3&skt=2026-01-15T06%3A18%3A19Z&ske=2026-01-16T06%3A18%3A19Z&sks=b&skv=2024-08-04&sig=cpPHVgXBsXQqzdH9L49j/nRWofA%2BPN%2By3RUY5Ytrmvg%3D')], output_format=None, quality=None, size=None, usage=None)

<https://oaidalleapiprodscus.blob.core.windows.net/private/org-oBzzAxWzofVRTZNu0vyqtv5g/user-3YLGCPD3BaT0GloeVuBPE3Y9/img-JQ9wMHFwAJG2bIn3ImD0orv0.png?st=2026-01-15T05%3A18%3A19Z&se=2026-01-15T07%3A18%3A19Z&sp=r&sv=2024-08-04&sr=b&rscd=inline&rsct=image/png&skoid=0ec745ad-3d9b-4d43-86db-87a2c9b420f1&sktid=a48cca56-e6da-484e-a814-9c849652bcb3&skt=2026-01-15T06%3A18%3A19Z&ske=2026-01-16T06%3A18%3A19Z&sks=b&skv=2024-08-04&sig=cpPHVgXBsXQqzdH9L49j/nRWofA%2BPN%2By3RUY5Ytrmvg%3D>



```
In [9]: prompt1 = "beautiful landscape with a tilt-shift lens" # 소형화된 시야
        prompt2 = "beautiful landscape with a 360 panorama"   # 360도 파노라마 뷰
        n = 1
        size = "512x512"

        kwargs1 = {
            "prompt" : prompt1,
            "n" : n,
            "size" : size,
```

```

}

kwargs2 = {
    "prompt" : prompt2,
    "n" : n,
    "size" : size,
}

im1 = client.images.generate(**kwargs1)
im2 = client.images.generate(**kwargs2)

for i in range(n):
    print(im1.data[i].url)
    display(Image(url=im1.data[i].url))
    print(im2.data[i].url)
    display(Image(url=im2.data[i].url))

```

<https://oaidalleapiprodscus.blob.core.windows.net/private/org-oBzzAxWzofVRTZNu0vyqtv5g/user-3YLGCPD3BaT0GloeVuBPE3Y9/img-AZTPypWldog3ovhPmoyNvRD7.png?st=2026-01-15T05%3A18%3A30Z&se=2026-01-15T07%3A18%3A30Z&sp=r&sv=2024-08-04&sr=b&rscd=inline&rsct=image/png&skoid=52f8f7b3-ca8d-4b21-9807-8b9df114d84c&sktid=a48cca56-e6da-484e-a814-9c849652bcb3&skt=2026-01-15T06%3A18%3A30Z&ske=2026-01-16T06%3A18%3A30Z&sks=b&skv=2024-08-04&sig=wtf5WgXR5eIndIr3u3Nxu1RDpp9TiL0BACjql/kYa90%3D>



<https://oaidalleapiprodscus.blob.core.windows.net/private/org-oBzzAxWzofVRTZNu0vyqtv5g/user-3YLGCPD3BaT0GloeVuBPE3Y9/img-n74Ceyid648q3gHsaXNCxlvX.png?st=2026-01-15T05%3A18%3A40Z&se=2026-01-15T07%3A18%3A40Z&sp=r&sv=2024-08-04&sr=b&rscd=inline&rsct=image/png&skoid=0ec745ad-3d9b-4d43-86db-87a2c9b420f1&sktid=a48cca56-e6da-484e-a814-9c849652bcb3&skt=2026-01-15T06%3A09%3A32Z&ske=2026-01-16T06%3A09%3A32Z&sks=b&skv=2024-08-04&sig=sfERRJPY9pEKGQLdYfsEwf1nZ6aHLNpatBSPKo6/zNA%3D>



12-6 여러 프롬프트 조합으로 다양한 랜덤 이미지 만들기

```
In [10]: import os
from openai import OpenAI
import random
import time
from IPython.display import Image, display
```

```
colors = ["Blue", "Red", "Green", "Yellow", "Purple", "Pink", "Orange",
          "Black", "White", "Gray", "Red and Green", "Yellow and Purple",
          "Orange and Blue", "Black and White", "Pink and Teal",
          "Brown and Lime", "Maroon and Violet", "Silver and Crimson",
          "Beige and Fuchsia", "Gold and Azure", "Cyan and Magenta",
          "Lime and Maroon and Violet", "Crimson and Silver and Gold",
          "Azure and Beige and Fuchsia", "Magenta and Cyan and Teal",
          "Pink and Teal and Lime", "Yellow and Purple and Maroon",
          "Orange and Blue and Violet", "Black and White and Silver",
          "Fade to Black", "Fade to White", "Fade to Gray", "Fade to Red",
          "Fade to Green", "Fade to Blue", "Fade to Yellow", "Fade to Purple",
          "Fade to Pink", "Fade to Orange", "Gradient of Red and Green",
          "Gradient of Yellow and Purple", "Gradient of Orange and Blue",
          "Gradient of Black and White", "Gradient of Pink and Teal",
          "Gradient of Brown and Lime", "Gradient of Maroon and Violet",
          "Gradient of Silver and Crimson", "Gradient of Beige and Fuchsia",
          "Gradient of Gold and Azure", "Gradient of Cyan and Magenta"]

resolution = ["2 bit colors", "4 bit colors", "8 bit colors", "16 bit colors",
              "24 bit colors", "4k resolution", "HDR", "8K resolution",
              "a million colors", "a billion colors"]

angles = ["Extreme close-up", "close-up", "medium shot", "long shot", "extreme long shot",
          "high angle", "overhead view", "aerial view", "tilted frame", "dutch angle",
          "over-the-shoulder shot", "drone view", "panning shot", "tracking shot",
          "dolly shot", "zoom shot", "handheld shot", "crane shot", "low angle",
          "reverse angle", "point-of-view shot", "split screen", "freeze frame",
          "flashback", "flash forward", "jump cut", "fade in", "fade out"]

lens = ["high-resolution microscopy", "microscopy", "macro lens", "pinhole lens",
        "knolling", "first person view", "wide angle lens", "lens distortion",
        "ultra-wide angle lens", "fisheye lens", "telephoto lens", "panorama",
        "360 panorama", "tilt-shift lens", "telescope lens", "lens flare",
        "Aperture: f/5.6, Shutter Speed: 1/250s, ISO: 400, Landscape photography, high-quality DSLR",
        "Aperture: f/8, Shutter Speed: 1/60s, ISO: 800, Street photography, low light conditions",
        "Aperture: f/11, Shutter Speed: 1/1000s, ISO: 1600, Sports photography, fast shutter speed",
        "Aperture: f/16, Shutter Speed: 2s, ISO: 100, Night photography, long exposure",
        "Aperture: f/2.8, Shutter Speed: 1/500s, ISO: 1600, Wildlife photography, high sensitivity, high ISO",
        "Aperture: f/4, Shutter Speed: 1/60s, ISO: 100, Portrait photography, shallow depth of field",
        "Aperture: f/5.6, Shutter Speed: 1/60s, ISO: 100, Macro photography, close-up shots",
        "Aperture: f/8, Shutter Speed: 1/15s, ISO: 100, Fine art photography, Kodak Gold 200 film color, 35mm",
```

```

        "Aperture: f/11, Shutter Speed: 4s, ISO: 200, Architectural photography, slow shutter speed, long exposure."]

light = ["Warm lighting", "Side lighting", "High-key lighting", "fluorescent lighting",
        "Harsh flash lighting", "Low-key lighting", "Flat lighting", "Even lighting",
        "Ambient lighting", "Colorful lighting", "Soft light", "Hard light",
        "Diffused light", "Direct light", "Indirect light", "Studio lighting",
        "Red and green lighting", "Flash photography", "Natural lighting",
        "Backlighting", "Edge lighting", "Cold", "Backlit", "Glow", "Neutral white",
        "High-contrast", "Lamp light", "Fireworks", "2700K light", "4800K light", "6500K light"]

filter = ["Kodachrome", "Autochrome", "Lomography", "Polaroid", "Instax", "Cameraphone",
        "CCTV", "Disposable camera", "Daguerrotype", "Camera obscura", "Double exposure",
        "Cyanotype", "Black and white", "Tri-X 400TX", "Infrared photography",
        "Bleach bypass", "Contact sheet", "Colour splash", "Solarised", "Anaglyph",
        "Instagram Clarendon filter", "Instagram Juno filter", "Instagram Ludwig filter",
        "Instagram Lark filter", "Instagram Gingham filter", "Instagram Lo-fi filter",
        "Instagram X-Pro II filter", "Instagram Aden filter", "Instagram Perpetua filter",
        "Instagram Reyes filter", "Instagram Slumber filter"]

lists = [colors, resolution, angles, lens, light, filter]

user_prompts = [
    "Happy Darth Vader smiling and waving at tourists in a museum of Star Wars memorabilia.",
    "Darth Vader rapping with 2Pac.",
    "Darth Vader playing the piano.",
    "Darth Vader playing the guitar.",
    "Darth Vader eating sushi.",
    "Darth Vader drinking a glass of milk.",
]

n = 2

for user_prompt in user_prompts:
    print(f"\nGenerating images for prompt: {user_prompt}")

    customizations = ""
    for lst in lists:
        if random.choice([True, False]):
            customizations += random.choice(lst) + ", "

    full_prompt = user_prompt + ", " + customizations

```

```
print("Using prompt:", full_prompt)

im = client.images.generate(
    prompt=full_prompt,
    n=n,
    size=size,
)

for idx, img in enumerate(im.data, start=1):
    print(f"Image {idx}")
    display(Image(url=img.url))
    time.sleep(1)

print("Finished generating images for prompt: " + user_prompt)
```

Generating images for prompt: Happy Darth Vader smiling and waving at tourists in a museum of Star Wars memorabilia.

Using prompt: Happy Darth Vader smiling and waving at tourists in a museum of Star Wars memorabilia., a billion colors, Ambient lighting, Double exposure,

Image 1



Image 2



Finished generating images for prompt: Happy Darth Vader smiling and waving at tourists in a museum of Star Wars memorabilia.

Generating images for prompt: Darth Vader rapping with 2Pac.

Using prompt: Darth Vader rapping with 2Pac., drone view, Diffused light, Black and white,
Image 1



Image 2



Finished generating images for prompt: Darth Vader rapping with 2Pac.

Generating images for prompt: Darth Vader playing the piano.

Using prompt: Darth Vader playing the piano., dolly shot, Instagram Clarendon filter,
Image 1



Image 2



Finished generating images for prompt: Darth Vader playing the piano.

Generating images for prompt: Darth Vader playing the guitar.
Using prompt: Darth Vader playing the guitar., high angle, Cold,
Image 1



Image 2



Finished generating images for prompt: Darth Vader playing the guitar.

Generating images for prompt: Darth Vader eating sushi.

Using prompt: Darth Vader eating sushi., Gradient of Gold and Azure, 2 bit colors, Side lighting, Instagram Ludwig filter, Image 1



Image 2



Finished generating images for prompt: Darth Vader eating sushi.

Generating images for prompt: Darth Vader drinking a glass of milk.

Using prompt: Darth Vader drinking a glass of milk., Gradient of Black and White, jump cut, Harsh flash lighting, Instagram Reels filter,

Image 1



Image 2



Finished generating images for prompt: DARTH VADER drinking a glass of milk.

DALL·E로 이미지 편집하기

```
In [11]: !curl -O https://raw.githubusercontent.com/LDJWJ/OpenAIGPTForPythonDevelopersFiles/main/image/ch13/mask.png
          !curl -O https://raw.githubusercontent.com/LDJWJ/OpenAIGPTForPythonDevelopersFiles/main/image/ch13/without_mask.png
          # 리눅스에서는 !wget
```

% Total	% Received	% Xferd	Average Speed	Time	Time	Time	Current
			Dload Upload	Total	Spent	Left	Speed
0	0	0	0	0	--:--:--	--:--:--	0
100	1368k	100	1368k	0	0	5355k	0
					--:--:--	--:--:--	5410k
% Total	% Received	% Xferd	Average Speed	Time	Time	Time	Current
			Dload Upload	Total	Spent	Left	Speed
0	0	0	0	0	--:--:--	--:--:--	0
1	1320k	1	16384	0	0	70430	0
				0:00:19	--:--:--	0:00:19	70620
100	1320k	100	1320k	0	0	3484k	0
				--:--:--	--:--:--	--:--:--	3494k

```
In [12]: client = OpenAI()

image = open("without_mask.png", "rb")
mask = open("mask.png", "rb")

prompt = "A group of people hiking in green forest between trees"
n = 1
size = "512x512"

kwargs = {
    "image": image,
    "mask": mask,
    "prompt": prompt,
    "n": n,
    "size": size,
}

response = client.images.edit(**kwargs)
image_url = response.data[0].url

print(image_url)
display(Image(url=image_url))
```

<https://oaidalleapiprodscus.blob.core.windows.net/private/org-oBzzAxWzofVRTZNU0vyqtv5g/user-3YLGCPD3BaT0GloeVuBPE3Y9/img-5R5qUHN1g5VMz5UMZXJ2iN2b.png?st=2026-01-15T05%3A20%3A02Z&se=2026-01-15T07%3A20%3A02Z&sp=r&sv=2024-08-04&sr=b&rscd=inline&rsct=image/png&skoid=92b997d6-f0dd-439d-8087-b26c74d72365&sktid=a48cca56-e6da-484e-a814-9c849652bcb3&skt=2026-01-15T06%3A16%3A37Z&ske=2026-01-16T06%3A16%3A37Z&skv=b&skv=2024-08-04&sig=IEbRlHWpJqlNFBBoxPwM%2B784E4S9IwhLNJj1b1s5tiss%3D>



다른 이미지에서 영감 얻기

이미지 변형 방법

```
In [13]: !curl -O https://raw.githubusercontent.com/LDJWJ/OpenAIGPTForPythonDevelopersFiles/main/image/ch14_dalle/ori_face_image.png
!curl -O https://raw.githubusercontent.com/LDJWJ/OpenAIGPTForPythonDevelopersFiles/main/image/ch14_dalle/dog01.png
```

```
!curl -O https://raw.githubusercontent.com/LDJWJ/OpenAIGPTForPythonDevelopersFiles/main/image/ch14_dalle/dog02.png
```

% Total	% Received	% Xferd	Average Speed	Time	Time	Time	Current
			Dload Upload	Total	Spent	Left	Speed
0	0	0	0	0	--:--:--	--:--:--	0
100	672k	100	672k	0	--:--:--	--:--:--	2732k
% Total	% Received	% Xferd	Average Speed	Time	Time	Time	Current
			Dload Upload	Total	Spent	Left	Speed
0	0	0	0	0	--:--:--	--:--:--	0
86	371k	86	320k	0	--:--:--	--:--:--	1103k
100	371k	100	371k	0	--:--:--	--:--:--	1276k
% Total	% Received	% Xferd	Average Speed	Time	Time	Time	Current
			Dload Upload	Total	Spent	Left	Speed
0	0	0	0	0	--:--:--	--:--:--	0
100	816k	100	816k	0	--:--:--	--:--:--	2766k

```
In [14]: client = OpenAI()

image = open("ori_face_image.png", "rb")
n = 3
size = "512x512"

kwargs = {
    "image": image,
    "n": n,
    "size": size
}

response = client.images.create_variation(**kwargs)
print(response)
```

```
ImagesResponse(created=1768458014, background=None, data=[Image(b64_json=None, revised_prompt=None, url='https://oaidalleapiprodscus.blob.core.windows.net/private/org-oBzzAxWzofVRTZNu0vyqtv5g/user-3YLGCPD3BaT0GloeVuBPE3Y9/img-mFswJnL4vpHhBe6vQiiOeFNT.png?st=2026-01-15T05%3A20%3A14Z&se=2026-01-15T07%3A20%3A14Z&sp=r&sv=2024-08-04&sr=b&rscd=inline&rsct=image/png&skoid=d505667d-d6c1-4a0a-bac7-5c84a87759f8&sktid=a48cca56-e6da-484e-a814-9c849652bcb3&skt=2026-01-15T06%3A20%3A14Z&ske=2026-01-16T06%3A20%3A14Z&skv=2024-08-04&sig=Q6waIMam1jiEjSpnlIok5iUPHnzNbBWdjLA7zxvjPzA%3D'), Image(b64_json=None, revised_prompt=None, url='https://oaidalleapiprodscus.blob.core.windows.net/private/org-oBzzAxWzofVRTZNu0vyqtv5g/user-3YLGCPD3BaT0GloeVuBPE3Y9/img-AQYncioPusyfkPoLbV6pb8c6.png?st=2026-01-15T05%3A20%3A14Z&se=2026-01-15T07%3A20%3A14Z&sp=r&sv=2024-08-04&sr=b&rscd=inline&rsct=image/png&skoid=d505667d-d6c1-4a0a-bac7-5c84a87759f8&sktid=a48cca56-e6da-484e-a814-9c849652bcb3&skt=2026-01-15T06%3A20%3A14Z&ske=2026-01-16T06%3A20%3A14Z&skv=2024-08-04&sig=WGhVrIDBeV9y6jr4LoMH6jisUktTbHTvWdxEzRnbY6M%3D'), Image(b64_json=None, revised_prompt=None, url='https://oaidalleapiprodscus.blob.core.windows.net/private/org-oBzzAxWzofVRTZNu0vyqtv5g/user-3YLGCPD3BaT0GloeVuBPE3Y9/img-j46Z05jjlDkCpgwsyd39yGZC.png?st=2026-01-15T05%3A20%3A14Z&se=2026-01-15T07%3A20%3A14Z&sp=r&sv=2024-08-04&sr=b&rscd=inline&rsct=image/png&skoid=d505667d-d6c1-4a0a-bac7-5c84a87759f8&sktid=a48cca56-e6da-484e-a814-9c849652bcb3&skt=2026-01-15T06%3A20%3A14Z&ske=2026-01-16T06%3A20%3A14Z&skv=2024-08-04&sig=gd8gF3fhgdG7LQKAhaOxcv6dNl/5zbWsdN4bG06u/y4%3D')], output_format=None, quality=None, size=None, usage=None)
```

```
In [15]: image = open("ori_face_image.png", "rb")
n = 3
size = "512x512"

kwargs = {
    "image": image,
    "n": n,
    "size": size
}

response = client.images.create_variation(**kwargs)
urls = response.data

for i in range(n):
    img_url = urls[i].url
    display(Image(url=img_url))
```








```
In [16]: image = open("dog01.png", "rb")
n = 2
size = "512x512"

kwargs = {
    "image": image,
    "n": n,
    "size": size
}
```

```
response = client.images.create_variation(**kwargs)
urls = response.data

for i in range(n):
    img_url = urls[i].url
    display(Image(url=img_url))
```





```
In [17]: image = open("dog02.png", "rb")
n = 2
size = "512x512"

kwargs = {
    "image": image,
    "n": n,
    "size": size
}
```



```
response = client.images.create_variation(**kwargs)
urls = response.data

for i in range(n):
    img_url = urls[i].url
    display(Image(url=img_url))
```



